

Fire & Smoke Detection

Final results report: every model trained, every measured number, and what each experiment showed
06 October 2026

Two models, one architecture (DINOv3 ViT-S/16 backbone + Faster R-CNN detector + image-level scene classifier). **Model 1** (640 px, two-stage training) is the proven one: zero false alarms on real industrial footage. **Model 6** (Model 1 fine-tuned at 896 px with mosaic augmentation) has the best accuracy and fixes small, distant fires; its false-alarm check on site footage is still to be run.

Headline number (test split, 4,306 images)	Value
Model 1: box detection, mAP@0.5 (smoke / fire)	0.7207 (0.814 / 0.628)
Model 1: fire/smoke images detected at 0.70% false alarms (conf 0.90)	87.4%
Model 1: fire/smoke images detected at 1.20% false alarms (conf 0.85)	92.3%
Model 1: false alarms on real industrial footage (54.5 min, 3 cameras, conf ≥ 0.75)	0
Model 6: mAP@0.5 (smoke / fire)	0.7291 (0.802 / 0.656)
Model 6: AP on tiny fires (<16 px), vs Model 1	0.385 vs 0.251
Fires alarmed on the video benchmark (conf 0.80): Model 6 / Model 1	7 of 7 / 5 of 7
Best backbone, frozen-feature comparison	DINOv3 (ahead of MobileNetV3 and ResNet-18)
Best light model	MobileNetV3-L: -2.3 mAP, about 2.5x faster on CPU

What was tried and what it showed

- **Two-stage DINOv3 training** (frozen backbone, then last 2 blocks unfrozen) improved the model: val 0.7312 -> 0.7406, test 0.7196 -> 0.7261 (both under the earlier test-time settings, see section 2).
- **Backbone comparison** with everything else fixed: DINOv3 0.7196 > MobileNetV3 0.6968 > ResNet-18 0.6477 on test. The choice of DINOv3 is supported by measurement.
- **4x more fire data** (DFS + FASDD, 14,792 fire images, with 1,931 leaked duplicates removed) left the test score essentially unchanged: 0.7222 vs 0.7207.
- **Small, distant fires** were the real weakness on CCTV video. Training at 896 px with mosaic raised tiny-fire AP from 0.251 to 0.385 and caught all 7 fires of a video benchmark. Pasting synthetic flames into training images was tried and rejected: it tripled false alarms.
- **Box mAP is limited by label consistency**, not by the model: AP drops by half when the box overlap requirement goes from 50% to 75%, and D-Fire itself labels the same smoke column as smoke in one frame and fire in the next.

1. The system and the data

Input frames are letterboxed to 640 x 640. A DINOv3 Vision Transformer (self-supervised, 21.6M parameters) produces features at four scales (strides 8/16/32/64); a Faster R-CNN head proposes and classifies boxes as *smoke* or *fire*; a second, image-level head reads the same features plus seven hand-crafted glow statistics and outputs P(smoke), P(fire) for the whole frame, so a flame hidden behind equipment can still be flagged from the light it casts. On video, an alarm requires 6 detections within 15 frames, which filters one-frame flicker.

Split (D-Fire, fixed for every experiment)	Images	Negatives	Smoke boxes	Fire boxes
Train	13,776	6,223	7,660	9,659
Validation	3,445	1,610	1,890	2,155
Test (never used for training or selection)	4,306	2,005	2,311	2,878

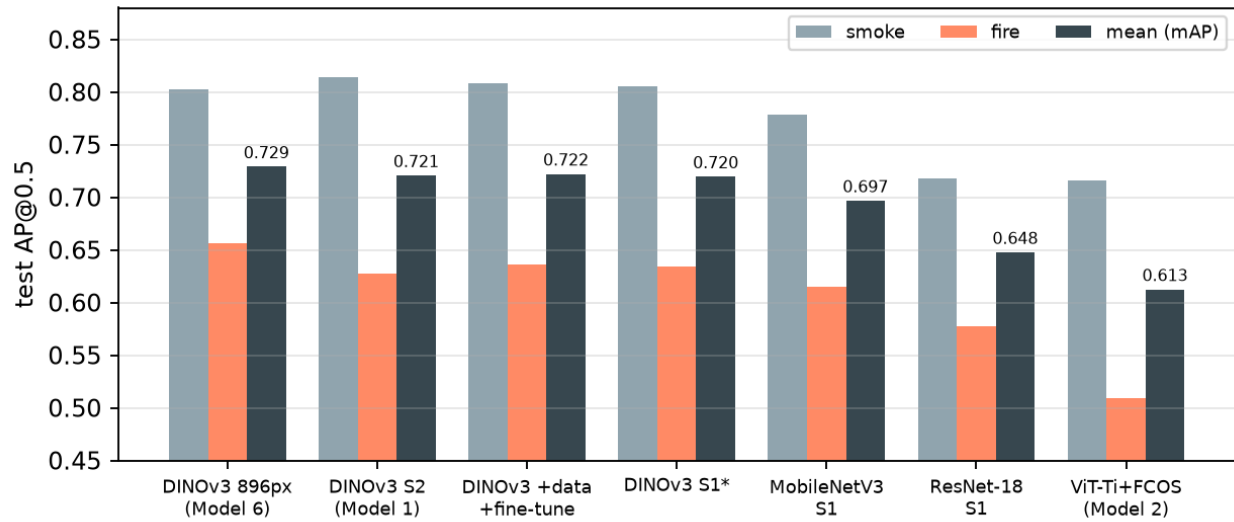
Training: AdamW, lr 1e-4 (5e-5 when unfreezing), backbone at 5% of the head learning rate, 500-step warmup then cosine decay, batch 8, mixed precision, early stopping on validation mAP@0.5; low-light, IR/grey, occlusion, crop, colour-jitter and blur augmentation. All runs on Kaggle T4 GPUs.

2. Every model trained

#	Model	Backbone state	Data	Epochs	Best val	Test mAP
1	DINOv3 ViT-S + Faster R-CNN, stage 1	frozen	D-Fire	40	0.7312	0.7196*
2	DINOv3 ViT-S + Faster R-CNN, stage 2 (Model 1)	last 2 blocks	D-Fire	15	0.7406	0.7207
3	DINOv3 ViT-Ti + FCOS (Model 2, light)	frozen	D-Fire	45	0.6220	0.6125
4	MobileNetV3-L + Faster R-CNN, stage 1	frozen	D-Fire	40	0.7056	0.6968
5	ResNet-18 + Faster R-CNN, stage 1	frozen	D-Fire	40	0.6567	0.6477
6	DINOv3, stage 3 from #2	last 2 blocks	D-Fire+DFS+FASDD	12	0.7177	-
7	DINOv3, fine-tune from #6 (epoch 4)	last 2 blocks	D-Fire	4	0.7251	0.7222
8	DINOv3 from #2, 896 px, mosaic + flame paste	last 2 blocks	D-Fire	6	0.7237	0.7190
9	DINOv3 from #2, 896 px, mosaic (Model 6)	last 2 blocks	D-Fire	6	0.7349	0.7291

*Validation picks the epoch; test is reported once per model. #7's best-validation epoch (epoch 1) scored 0.7128 on test; epoch 4 is shown because it was better on both splits' alarm metrics. * Test-time settings were changed for speed part-way through the project (300 instead of 1000 proposals, at most 20 instead of 50 boxes per image). Stage 2 scores 0.7261 under the earlier settings and 0.7207 under the current ones; every other row uses the current ones except stage 1, which is still to be re-evaluated and should read about 0.005 lower.*

3. Test-split results

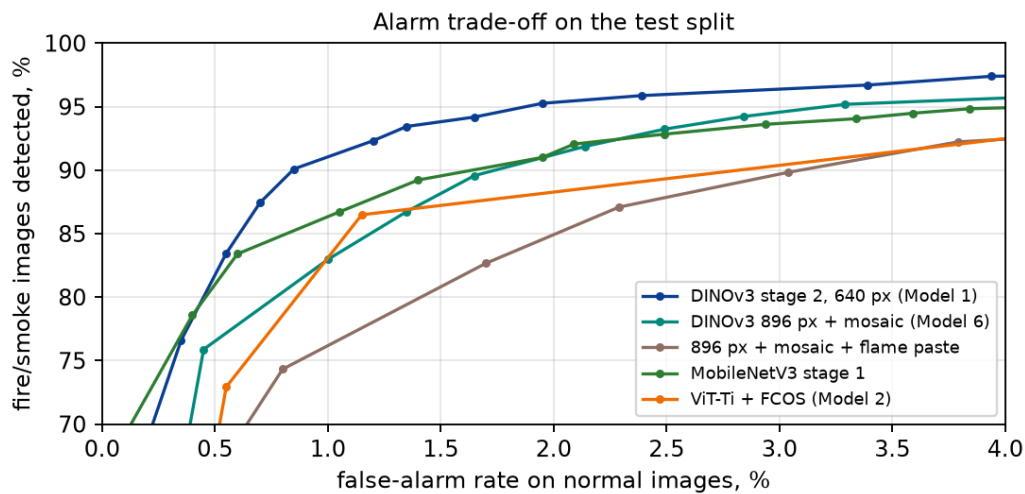


AP@0.5 per class on the 4,306-image test split. Model 6 has the best mAP and fire AP; Model 1 the best smoke AP. * earlier test-time settings (section 2).

Model	Smoke AP	Fire AP	mAP@0.5	mAP@0.75	mAP@.5:.95
DINOv3 896 px + mosaic (Model 6)	0.8023	0.6559	0.7291	0.3152	0.3612
DINOv3 stage 2 (Model 1)	0.8137	0.6278	0.7207	0.3211	0.3629
DINOv3 + data + fine-tune	0.8082	0.6362	0.7222	0.3338	0.3665
DINOv3 896 px + mosaic + paste	0.8021	0.6360	0.7190	0.3134	0.3563
DINOv3 stage 1*	0.8050	0.6342	0.7196	0.3252	0.3627
MobileNetV3-L stage 1	0.7786	0.6150	0.6968	0.2714	0.3334
ResNet-18 stage 1	0.7177	0.5778	0.6477	0.2426	0.3046
ViT-Ti + FCOS (Model 2)	0.7158	0.5092	0.6125	0.2205	0.2854

4. Alarm behaviour: the number that decides deployment

A deployed system raises an alarm per frame, not per box. The test sweep below counts, for each confidence threshold, how many *normal* images produced any box (false-alarm rate) and how many fire/smoke images were detected.



Model (test split)	Threshold	FP images	FP rate	Detected (any)	Fire	Smoke
DINOv3 stage 2 (Model 1)	0.90	14	0.70%	87.4%	77.7%	83.7%
DINOv3 stage 2 (Model 1)	0.85	24	1.20%	92.3%	85.6%	89.8%
DINOv3 896 px + mosaic (Model 6)	0.90	27	1.35%	86.7%	78.6%	82.3%
DINOv3 896 px + mosaic (Model 6)	0.93	20	1.00%	83.0%	71.7%	78.0%
DINOv3 + data + fine-tune	0.90	14	0.70%	86.8%	78.8%	82.6%
DINOv3 + data + fine-tune	0.85	24	1.20%	91.6%	87.0%	88.6%
DINOv3 stage 1	0.90	14	0.70%	84.8%	75.9%	81.2%
MobileNetV3-L stage 1	0.90	12	0.60%	83.4%	73.3%	79.5%
ResNet-18 stage 1	0.90	13	0.65%	72.9%	61.4%	68.0%

On still photos Model 1 keeps the best alarm trade-off: at about 1% false alarms it detects 90.1% (conf 0.88) against 83.0% for Model 6 (conf 0.93). At 896 px the model also sees more small bright objects in normal scenes.

Real industrial footage

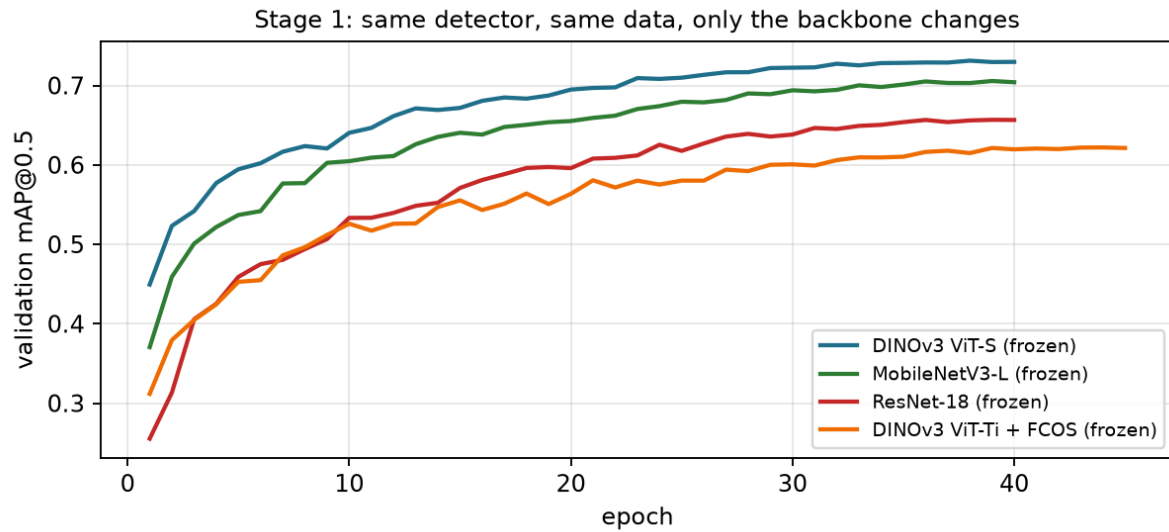
Model 1 was also run on 54.5 minutes of normal operation from three fixed industrial cameras (confidential footage, no fire present), sampled at 1 fps with the 6-of-15 confirmation rule. This measures false alarms only.

Confidence threshold	False alarms	Per hour
0.50	19	20.9
0.60	9	9.9
0.70	3	3.3
0.75	0	0.0
0.85	0	0.0
0.90	0	0.0

Zero false alarms from 0.75 upward, so the 0.85-0.90 operating point has margin. What triggered below that: bright dusty haze (as smoke) and a worker's yellow hard hat (as fire), both stationary, which is the case temporal confirmation cannot filter. They are the first candidates for hard-negative training.

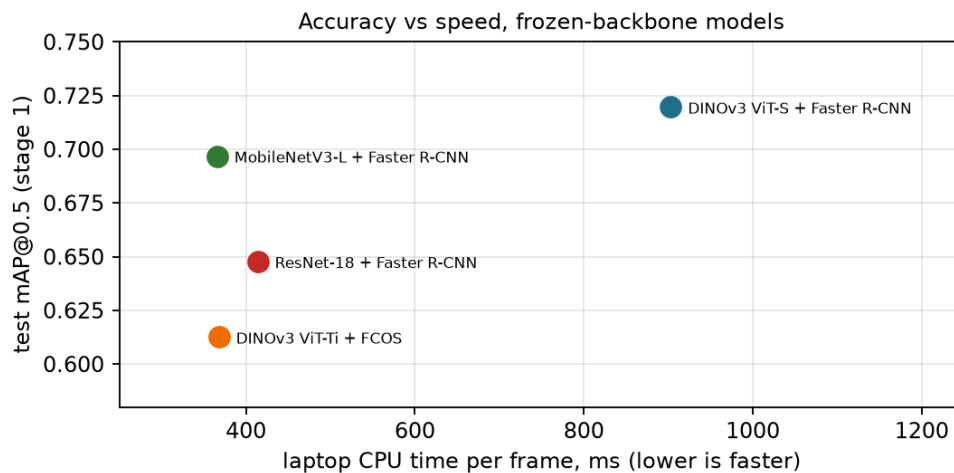
5. Backbone comparison: is DINOv3 worth it?

The same detector was trained on two ImageNet CNN backbones, each on its own code branch. Only the backbone changed: Faster R-CNN head, scene head, glow prior, augmentation, optimiser and the 40-epoch frozen-backbone schedule are identical. With the backbone frozen, the comparison tests the quality of the pretrained features directly.



Backbone (stage 1)	Params	Val mAP	Test mAP	Test fire AP	Detected @0.90	CPU ms/frame
DINOv3 ViT-S	39.3M	0.7312	0.7196	0.6342	84.8%	894-912
MobileNetV3-L	20.0M	0.7056	0.6968	0.6150	83.4%	346-385
ResNet-18	27.9M	0.6567	0.6477	0.5778	72.9%	414
DINOv3 ViT-Ti + FCOS	6.9M	0.6220	0.6125	0.5092	n/a*	338-398

CPU times: project laptop, batch 1, 640 px, fp32, median of 15 runs; ranges where two sessions differed. * Model 2's FCOS head compresses scores (max about 0.75), so 0.90 is not a usable threshold for it; at a matched 0.7% false-alarm rate it detects 76.3%. DINOv3 stage 1 was evaluated under the earlier test-time settings and the CNNs under the current ones (section 2), so its lead is about 0.5 point smaller than shown; it remains ahead.



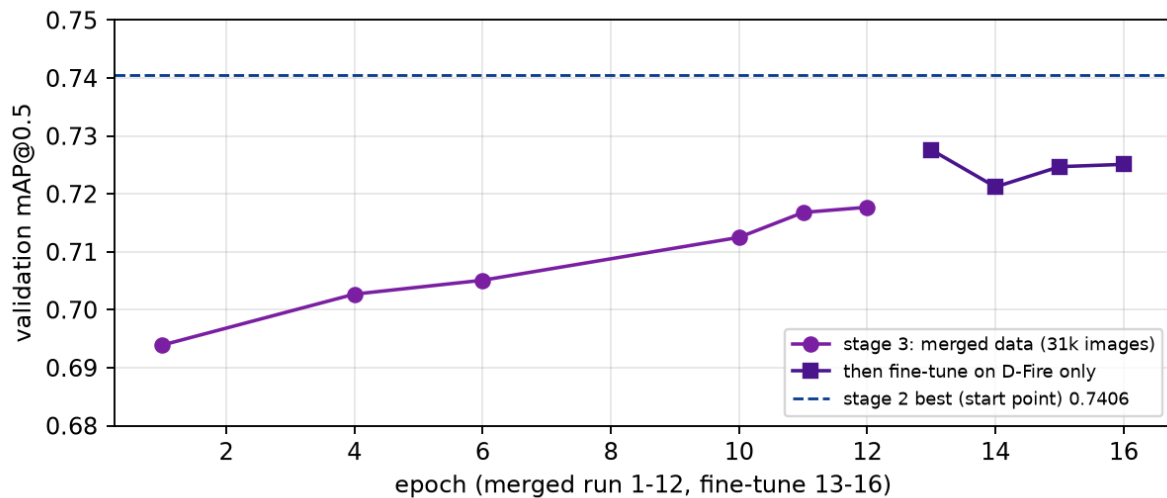
- **DINOv3 wins on accuracy** on both splits and both classes: +2.3 mAP over MobileNetV3 and +7.2 over ResNet-18 on test.
- **MobileNetV3 is the best light option**: at the speed of Model 2 it scores +8.4 mAP, and it is close to DINOv3 at the alarm threshold.
- **ResNet-18 is dominated**: slower than MobileNetV3 and clearly less accurate.

6. Experiment: 4x more fire data

Fire appeared in half as many training images as smoke (3,819 vs 6,778), and fire AP was the weakest number. Two public datasets were added to the training split only, through a merge tool that maps class names explicitly and removes near-duplicates of every existing image (perceptual hash, confirmed pixel by pixel).

	DFS	FASDD_CV
Fire images added	6,792	8,000 (capped)
Negatives added	1,008 hard negatives (lamps, sunsets...)	
Duplicates of D-Fire test images removed	34	341
Duplicates of D-Fire val / train removed	26 / 87	290 / 1,153

Training set 13,776 -> 31,076 images. Without the duplicate check, 375 test images would have entered training and inflated the test score.

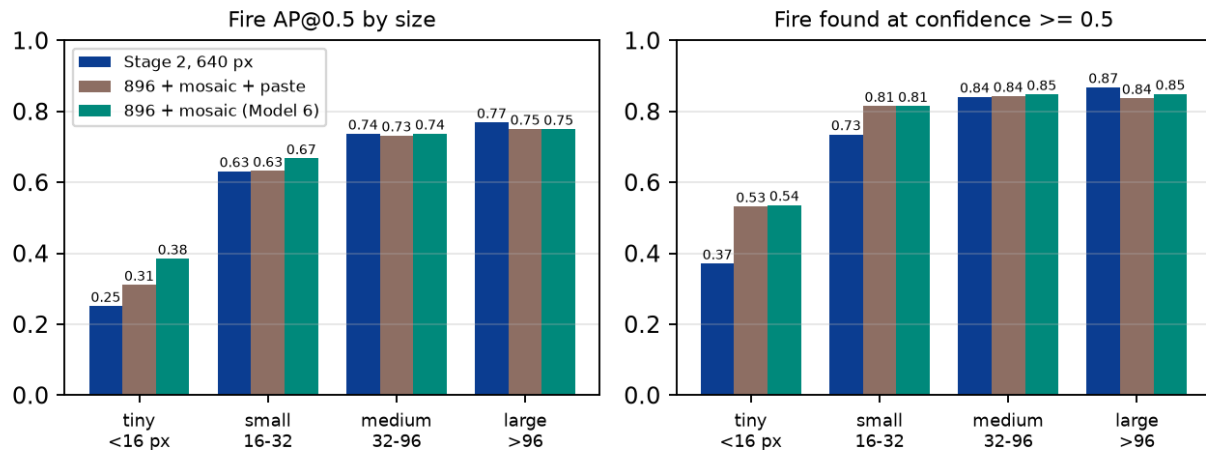


Test split	Smoke AP	Fire AP	mAP@0.5	Detected @0.90 (FP images)
DINOv3 stage 2 (start point)	0.8137	0.6278	0.7207	87.4% (14)
After merged data + D-Fire fine-tune	0.8082	0.6362	0.7222	86.8% (14)

Result. The extra data did not improve the D-Fire test score; the best checkpoint ties stage 2 (0.7222 vs 0.7207, same number of false-alarm images at 0.90). Validation first dropped by 4.7 points on the mixed data - mostly smoke - and only partly recovered after a D-Fire-only fine-tune. The external datasets draw fire and smoke boxes differently, and the D-Fire benchmark rewards D-Fire's own convention. Whether the broader data helps on unfamiliar industrial scenes is not measurable on D-Fire and is the next test.

7. Small and distant fires

Two industrial CCTV clips (1920 x 1080) with fires edited in showed the weakness the photo benchmark hides: large and medium fires scored 0.97+, but fires about 45 px wide were *seen and not confirmed* - scores of 0.61-0.86, below the alarm threshold. Letterboxing a 1920 px frame to 640 px shrinks such a flame to about 13 px. The tracker was ruled out (consecutive boxes overlap at IoU 0.79-0.84). To measure this, evaluation now reports accuracy by object size, and a fixed video benchmark scores every fire separately.



Test split, fire class, by object size (square root of box area, in pixels of a 640 px input).

Test split	Model 1 (640 px)	896 px + mosaic + paste	Model 6 (896 px + mosaic)
mAP@0.5	0.7207	0.7190	0.7291
Fire AP	0.6278	0.6360	0.6559
Tiny fire AP (<16 px)	0.251	0.311	0.385
Small fire AP (16-32 px)	0.631	0.632	0.667
Tiny fires found at conf >= 0.5	37%	53%	54%
False-alarm images at conf 0.90 (of 2,005)	14	46	27
Scene head: false 'fire' rate	1.4%	6.6%	1.6%

Three fine-tunes from Model 1, six epochs each. Training at 896 px with mosaic augmentation (2x2 or 3x3 images tiled into one frame, so objects appear at 1/2-1/3 size) raised tiny-fire AP by 13 points and gave the best mAP of the project. Adding *flame pasting* (real flames cut from training boxes and blended in at 8-40 px) did not help further and tripled false alarms (46 images against 14); the model learned that a small warm blob is fire. A repeat of the paste run reproduced it (44 images), and switching paste off removed most of the damage. Pasting was dropped.

Video benchmark

Seven fires in two clips; an alarm needs 6 detections within 15 processed frames (every 5th frame) at confidence 0.80. Each cell shows the median fire score inside the fire's time window and the delay from the fire appearing to the alarm.

Fire	Size	Model 1, 640 px	Model 1, 640+960 px	Model 6, 896 px	Paste model, 896 px
Clip 1, medium, centre	147 px	0.98 / +0.8 s	+0.8 s	0.99 / +0.8 s	0.99 / +0.7 s
Clip 2, large, centre	140 px	0.98 / +1.0 s	+1.0 s	0.99 / +1.0 s	0.99 / +0.9 s
Clip 2, medium, centre	97 px	0.97 / +0.8 s	+0.8 s	0.92 / +0.8 s	0.99 / +0.8 s
Clip 2, distant, upper (1st)	43 px	0.63 / missed	+0.7 s	0.84 / +2.0 s	0.98 / +0.7 s
Clip 2, distant, upper (2nd)	44 px	0.61 / missed	+1.0 s	0.88 / +1.0 s	0.99 / +0.7 s
Clip 2, small, right	49 px	0.86 / +1.3 s	+0.8 s	0.91 / +1.1 s	0.98 / +0.8 s
Clip 2, small, lower left	53 px	0.80 / +2.0 s	+2.0 s	0.85 / +1.3 s	0.98 / +0.8 s
Fires alarmed / false alarms		5 of 7 / 0	7 of 7 / 0	7 of 7 / 0	7 of 7 / 0
Passes per frame / laptop CPU s		1 / 0.9	2 / 3.2	1 / 2.0	1 / 2.0

Model 6 at higher thresholds: 6 of 7 fires at 0.90, 3 of 7 at 0.93. The 960 px pass of the two-size setup was run on the 75-98 s section of clip 2 only.

- **The resolution gain is real.** Model 6 never saw an edited flame in training, and the distant fire rose from 0.61-0.63 to 0.84-0.88; all seven fires alarm in a single pass.
- **The paste model's 0.98 is partly an artefact:** it was trained on pasted flames and the test fires are pasted flames. The same fires score 0.84-0.91 without that training.
- **Limits of this benchmark.** The fires are edited in, so they do not light their surroundings; fire windows were derived from detections and await confirmation against the edit timeline; and two clips cannot measure false alarms. A real small flame filmed on the site camera is the decisive test, and Model 6 has not yet been run on the 54.5-minute site footage.

8. Why box mAP levels off around 0.72-0.74

mAP@0.5 counts a detection as correct only if its box overlaps the annotator's box by at least 50%. Tightening that to 75% roughly halves every score, which means the model usually finds the object but disagrees with the annotator about its edges - and fire and smoke have no sharp edges.

Model 1, test split	Overlap $\geq$ 50%	Overlap $\geq$ 75%
Smoke	0.814	0.407
Fire	0.628	0.235
Mean	0.721	0.321



Two D-Fire training frames taken 10 seconds apart (timestamps visible). The same thin smoke column is labelled **smoke** (blue, left) and **fire** (red, right). A model cannot score above the consistency of its labels.

- Fire is scored lower than smoke even though it is easier to see: several flames are boxed as one region by some annotators and one box per flame by others, and a correct single box then counts as one hit and several misses.
- Training has levelled off: stage 2 gained about 0.001 mAP per epoch at the end, and 4x more data did not move the test score.
- At the frame level, which is what triggers an alarm, Model 1 already detects 92.3% of fire/smoke images at a 1.2% false-alarm rate.

9. Conclusions and next steps

- **Model 1 (DINOV3 stage 2, 640 px) is the proven model:** best alarm trade-off on test photos and zero false alarms on real industrial footage at the operating thresholds.
- **Model 6 (896 px + mosaic) is the candidate to replace it:** best mAP, best small-fire accuracy, 7 of 7 video fires in one pass. It costs about 2.2x the compute and false-alarms on about twice as many normal test photos at the same threshold, so the site-footage check decides.
- **DINOV3 is the right backbone:** it beat both ImageNet CNNs with everything else held fixed.
- **MobileNetV3 is the candidate for scaling to many cameras:** close to DINOV3 at the alarm threshold at about 2.5x the speed. A MobileNetV3 + FCOS version would add batched multi-camera export.
- **Next: hard negatives.** The known false triggers are fire-coloured objects - yellow and orange hard hats, hi-vis clothing, bright haze. Adding such images as verified negatives (public safety-helmet datasets, the 1,008 DFS 'other' images already prepared, and site frames) is the direct way to bring Model 6's false alarms down to Model 1's level.
- **Then: a real-fire test on the site camera** (a small controlled flame at several distances), and re-evaluating stage 1 under the current test-time settings so every number in this report is measured identically.